Γ, ${}_0F_1$ and J_0

A short introduction to everything `list-and-bag.tex` assumes, starting from $n!$

Four things are used without explanation in `list-and-bag.tex`: the Gamma function, the reflection formula, the hypergeometric notation ${}_0F_1$, and the Bessel function J_0. This note introduces them in that order, because each one needs the one before it. Nothing here is original and nothing here is hard; the only claim the note makes for itself is that it is the shortest path to the four.

1 Γ: the factorial, for non-integers

1.1 The problem it solves

We want a function that agrees with $n!$ at the positive integers but is defined — and smooth — in between. Interpolating a sequence is not by itself a well-posed problem: infinitely many smooth curves pass through the points $1, 1, 2, 6, 24, \ldots$. What picks one out is insisting on the *recursion* rather than just the values. Factorial satisfies

$$(n+1)! = (n+1)\cdot n!,$$

so we ask for a function f on $(0,\infty)$ with $f(1)=1$ and $f(x+1) = x\,f(x)$ for all $x>0$.

That is still not enough — one can perturb a solution by any function of period 1 — but it very nearly is.

Theorem 1.1 (Bohr–Mollerup)**.** *There is exactly one function $f:(0,\infty)\to(0,\infty)$ with*

$$f(1)=1, \qquad f(x+1)=xf(x), \qquad \log f \text{ convex.}$$

Log-convexity is the mild extra demand that rules out the periodic wobbles. The unique solution is called Γ, and it is given by an integral.

1.2 The definition

Definition 1.2. For $\operatorname{Re} z > 0$,

$$\Gamma(z) \;=\; \int_0^\infty t^{\,z-1}e^{-t}\,dt.$$

Two computations establish everything basic. First, integrating by parts with $u = t^z$ and $dv = -e^{-t}dt$:

$$\Gamma(z+1) = \int_0^\infty t^z e^{-t}\,dt = \underbrace{\big[-t^z e^{-t}\big]_0^\infty}_{=0} + z\int_0^\infty t^{z-1}e^{-t}\,dt = z\,\Gamma(z).$$

Second, $\Gamma(1) = \int_0^\infty e^{-t}dt = 1$. Together these give, by induction,

$$\boxed{\Gamma(n+1) = n! \quad \text{for } n\in\mathbb{N}.} \tag{1}$$

So Γ is the factorial with its argument shifted by one. The shift is an accident of history and is permanently annoying; the thing to memorise is (1).

Remark 1.3 (why the integral, and not some formula). Definition 1.2 is not a clever trick for computing $n!$ — it is *why* Γ *appears in this subject at all.* Read it right to left: the integral $\int_0^\infty t^n e^{-t} dt$ *produces* $n!$ out of a power. Whenever a construction divides by $n!$ and you want to undo that, this integral is the tool, and that is exactly the role it plays in `list-and-bag.tex` §4: the Borel transform $\frac{1}{1-x} = \int_0^\infty e^{-t} e^{xt} dt$ is (1) applied coefficient by coefficient.

1.3 The one value that matters here: $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Proposition 1.4. $\Gamma(\frac{1}{2}) = \displaystyle\int_{\mathbb{R}} e^{-u^2}\, du = \sqrt{\pi}.$

Proof. Put $t = u^2$ in Definition 1.2, so $dt = 2u\, du$ and $t^{-1/2} = u^{-1}$:

$$\Gamma(\tfrac{1}{2}) = \int_0^\infty t^{-1/2} e^{-t}\, dt = \int_0^\infty u^{-1} e^{-u^2}\, 2u\, du = 2\int_0^\infty e^{-u^2}\, du = \int_{\mathbb{R}} e^{-u^2}\, du.$$

That the last integral is $\sqrt{\pi}$ is the Gaussian integral, proved by squaring it and passing to polar coordinates. □

This is worth pausing on, because it is the whole hinge of the other note. $\Gamma(\frac{1}{2})$ *and the Gaussian integral are the same object.* The substitution $t = u^2$ is the only thing between them. So when `list-and-bag.tex` substitutes $-x^2$ into a generating function and integrates, it is not doing something to which Γ is incidentally relevant: it is evaluating Γ at a half-integer, by construction.

1.4 Outside the right half-plane

The integral of Definition 1.2 diverges for $\operatorname{Re} z \le 0$, but the recursion $\Gamma(z) = \Gamma(z+1)/z$ extends Γ to all of $\mathbb{C}$ except $z = 0, -1, -2, \ldots$, where it has simple poles. The useful consequence:

Proposition 1.5. $1/\Gamma$ *is entire — analytic on all of* $\mathbb{C}$*, with zeros exactly at* $0, -1, -2, \ldots$.

This is the fact that licenses the proof of `list-and-bag.tex` Theorem 5.1. There the coefficients of F_s are $\varphi(n) = \Gamma(n+1)^{-s}$, and Ramanujan's master theorem requires φ to be analytic at a *negative* argument, namely $\varphi(-\frac{1}{2}) = \Gamma(\frac{1}{2})^{-s}$. Had Γ itself been needed there instead of $1/\Gamma$, the pole structure would have obstructed it.

2 The reflection formula

2.1 Statement

Theorem 2.1 (Euler's reflection formula)**.** *For* $z \notin \mathbb{Z}$*,*

$$\Gamma(z)\,\Gamma(1-z) \;=\; \frac{\pi}{\sin \pi z}.$$

The name: $z \mapsto 1 - z$ reflects the complex plane in the vertical line $\operatorname{Re} z = \frac{1}{2}$, and the formula says Γ paired with its own reflection is elementary. Note the consistency check: the left side has poles at every integer (from $\Gamma(z)$ at $z \le 0$ and from $\Gamma(1-z)$ at $z \ge 1$), and so does $\pi/\sin \pi z$. **checked**

2.2 Where it comes from

A sketch, because the mechanism is the same integral that appears in the other note. The Beta integral is

$$B(a,b) \;=\; \int_0^1 t^{a-1}(1-t)^{b-1}\,dt \;=\; \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

which one proves by writing $\Gamma(a)\Gamma(b)$ as a double integral and changing variables. Substituting $t = x/(1+x)$ turns it into an integral over $(0,\infty)$:

$$\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \;=\; \int_0^\infty \frac{x^{a-1}}{(1+x)^{a+b}}\,dx.$$

Now take $b = 1-a$, so that $\Gamma(a+b) = \Gamma(1) = 1$ and the denominator is just $1+x$:

$$\Gamma(a)\Gamma(1-a) \;=\; \int_0^\infty \frac{x^{a-1}}{1+x}\,dx. \tag{2}$$

Evaluating (2) by contour integration (a keyhole around the cut on the positive axis) gives $\pi/\sin\pi a$, which is the theorem.

2.3 The case we need, and what it explains

Set $z = \frac{1}{2}$ in Theorem 2.1. Then $1-z = z$, $\sin\frac{\pi}{2} = 1$, and

$$\boxed{\Gamma(\tfrac{1}{2})^2 \;=\; \pi.} \tag{3}$$

Which we already knew from Proposition 1.4. The point is not the value but *which* identity produces it, and that $\frac{1}{2}$ is the fixed point of the reflection.

Remark 2.2 (the π in the List row *is* this formula). Put $a = \frac{1}{2}$ in (2) and substitute $x = u^2$:

$$\Gamma(\tfrac{1}{2})\Gamma(\tfrac{1}{2}) = \int_0^\infty \frac{x^{-1/2}}{1+x}\,dx = \int_0^\infty \frac{2\,du}{1+u^2} = \int_{\mathbb{R}} \frac{du}{1+u^2}.$$

So $\int_{\mathbb{R}} dx/(1+x^2) = \pi$ is not merely *equal to* $\Gamma(\frac{1}{2})^2$ — it is the reflection formula at $a = \frac{1}{2}$, written in a different variable. This is the content of `list-and-bag.tex` Remark 5.2: the exponent 2 in $\Gamma(\frac{1}{2})^2$ comes from reflection pairing $\Gamma(a)$ with $\Gamma(1-a)$, and the x^2 substitution is what sets $a = \frac{1}{2}$, the one place where those two Γ-values coincide and the product becomes a *square*. At any other exponent x^m one gets $\Gamma(\frac{1}{m})\Gamma(1-\frac{1}{m})$ — still elementary, but no longer a power of one number.

3 Hypergeometric notation: what ${}_0F_1$ means

3.1 Rising factorials

Definition 3.1. The *Pochhammer symbol* (rising factorial) is

$$(b)_n \;=\; b(b+1)(b+2)\cdots(b+n-1) \;=\; \frac{\Gamma(b+n)}{\Gamma(b)}, \qquad (b)_0 = 1.$$

The only instance needed below is $b = 1$:

$$(1)_n = 1\cdot 2\cdots n = n!. \tag{4}$$

3.2 The notation

Hypergeometric notation is a uniform way of naming power series whose coefficient *ratios* are rational in n. The general shape is

$$ {}_pF_q\left(\begin{matrix}a_1,\dots,a_p\\ b_1,\dots,b_q\end{matrix};z\right) \;=\; \sum_{n\ge 0}\frac{(a_1)_n\cdots(a_p)_n}{(b_1)_n\cdots(b_q)_n}\,\frac{z^n}{n!}. \tag{5} $$

Read it as: p rising factorials upstairs, q downstairs, and *always* a spare $n!$ in the denominator. When $p=0$ the upstairs list is empty and one writes a bare semicolon in its place: ${}_0F_q(\,;b_1,\dots,b_q;z)$. That is the only feature of the notation that looks like a typo and is not.

Example 3.2 ($p=q=0$)**.** ${}_0F_0(\,;\,;z)=\sum_n z^n/n!=e^z$. The exponential is the emptiest hypergeometric series.

Example 3.3 ($p=0$, $q=1$, $b_1=1$)**.** Using (4),

$$ {}_0F_1(\,;1;z) \;=\; \sum_{n\ge0}\frac{1}{(1)_n}\cdot\frac{z^n}{n!} \;=\; \sum_{n\ge0}\frac{z^n}{n!\cdot n!} \;=\; \sum_{n\ge0}\frac{z^n}{(n!)^2}. $$

So the single parameter $b_1=1$ is doing exactly one job: it contributes a second $n!$ to the denominator. Repeating it repeats that job, which is the whole reason the notation shows up in `list-and-bag.tex`:

Proposition 3.4. *With* $F_s(z)=\sum_{n\ge0}z^n/(n!)^s$,

$$ F_s(z) \;=\; {}_0F_{s-1}\big(\,;\underbrace{1,\dots,1}_{s-1};z\big). $$

In particular $F_1={}_0F_0=e^z$, *and* $F_2={}_0F_1(\,;1;z)$.

Proof. By (5) with $p=0$ and all $b_i=1$, the n-th coefficient is $1/\big((1)_n^{s-1}\,n!\big)=1/(n!)^{s-1}n!=1/(n!)^s$ using (4). □

Remark 3.5. $F_0(z)=\sum z^n=1/(1-z)$ is *not* of the form ${}_0F_{-1}$ — the notation starts at $s=1$. In hypergeometric terms F_0 is ${}_1F_0(1;\,;z)$: one parameter upstairs, $(1)_n=n!$, cancelling the obligatory $n!$ downstairs. The geometric series sits on the other side of the notation from the exponential, which is a small notational echo of the List/Bag distinction the other note is about.

4 J_0: the Bessel function, and why it is F_2

4.1 Definition

Definition 4.1. The Bessel function of the first kind of order zero is

$$ J_0(w) \;=\; \sum_{k\ge0}\frac{(-1)^k}{(k!)^2}\Big(\frac{w}{2}\Big)^{2k}. $$

It is the solution of $w^2y''+wy'+w^2y=0$ regular at the origin, and it turns up whenever a problem has circular symmetry — the vibration of a drumhead, the diffraction pattern of a round aperture. For present purposes none of that matters; only the series does.

4.2 The identification

Proposition 4.2. *$F_2(-x^2) = J_0(2x)$, and $F_2(z) = I_0(2\sqrt{z})$ where I_0 is the modified Bessel function (Definition 4.1 without the $(-1)^k$).*

Proof. Substitute $w = 2x$ in Definition 4.1: the $(w/2)^{2k}$ becomes x^{2k}, so

$$J_0(2x) = \sum_{k\geq 0} \frac{(-1)^k x^{2k}}{(k!)^2} = \sum_{k\geq 0} \frac{(-x^2)^k}{(k!)^2} = F_2(-x^2)$$

by Example 3.3. The halving in Definition 4.1 is precisely why the argument is $2x$ and not x. □

So the chain of names in `list-and-bag.tex` unwinds as:

$$F_2(z) = \sum \frac{z^n}{(n!)^2} \xrightarrow[\text{Prop. 3.4}]{} {}_0F_1(\,;1;z) \xrightarrow[\text{Prop. 4.2}]{} J_0(2x) \text{ at } z = -x^2.$$

Three notations, one series. The reason to mention the third at all is that J_0 has a classical integral that the other two do not advertise:

$$\int_0^\infty J_0(t)\,dt = 1 \qquad \Longrightarrow \qquad \int_{\mathbb{R}} J_0(2x)\,dx = 1, \tag{6}$$

which is the $s = 2$ case of the theorem, and equals $\Gamma(\frac{1}{2})^{2-2} = 1$. **checked**

4.3 The convergence caveat, concretely

For large real w,

$$J_0(w) \;\sim\; \sqrt{\frac{2}{\pi w}}\,\cos\!\Big(w - \frac{\pi}{4}\Big),$$

so J_0 oscillates with amplitude decaying only like $w^{-1/2}$. That is too slow for absolute convergence — $\int |J_0|$ diverges — but fast enough, with the sign changes, for (6) to hold conditionally. This is the first point in the family F_s at which the integral stops being an ordinary convergent integral, and it is why `list-and-bag.tex` Remark 5.3 damps by $e^{-\varepsilon x}$ rather than integrating directly.

Beyond $s = 2$ it gets worse in a specific way. The oscillation of ${}_0F_{s-1}(\,;1,\ldots,1;-x^2)$ is not in x but in $x^{2/s}$, so for $s \geq 3$ the function is not even *periodic* in x — there is no period to feed to an oscillatory quadrature rule, and the integral exists only in the regularised (Mellin/Abel) sense. The $s = 2$ case is the last one where a classical integral formula like (6) is available, which is why it is the last one verified numerically in the other note.

5 Where each of these is used

object	here	role in `list-and-bag.tex`
$\Gamma(n+1) = n!$	(1)	the divisor that distinguishes Bag from List
$\int_0^\infty t^n e^{-t} dt = n!$	Rem. 1.3	the Borel transform of §4 — the de-quotienting map
$\Gamma(\frac{1}{2}) = \sqrt{\pi}$	Prop. 1.4	the Gaussian integral; the Bag row
$1/\Gamma$ entire	Prop. 1.5	licenses $\varphi(-\frac{1}{2})$ in the proof of Thm. 5.1
$\Gamma(a)\Gamma(1-a) = \pi/\sin \pi a$	Thm. 2.1	the exponent 2 in $\Gamma(\frac{1}{2})^{2-s}$; Rem. 5.2
$a = \frac{1}{2}$ self-dual	(3), Rem. 2.2	why the *pair* container x^2 and no other power
${}_0F_{s-1}$	Prop. 3.4	a name for "divide by $n!$ exactly s times"
$J_0(2x) = F_2(-x^2)$	Prop. 4.2	makes $s = 2$ numerically checkable
$\int_0^\infty J_0 = 1$	(6)	the $s = 2$ value, $\Gamma(\frac{1}{2})^0$
$J_0 \sim w^{-1/2} \cos$	§4.3	where convergence stops, and why

Further reading

Artin, *The Gamma Function* (1931).
Thirty-nine pages, and still the best introduction. Theorem 1.1 is its spine.

Andrews, Askey, Roy, *Special Functions* (1999).
Chapters 1–2 for Γ, B and reflection; Chapter 4 for (5). The standard modern reference.

Whittaker and Watson, *A Course of Modern Analysis*.
Chapter XII for Γ, XVII for Bessel. Where to look when a formula is needed and not understood.